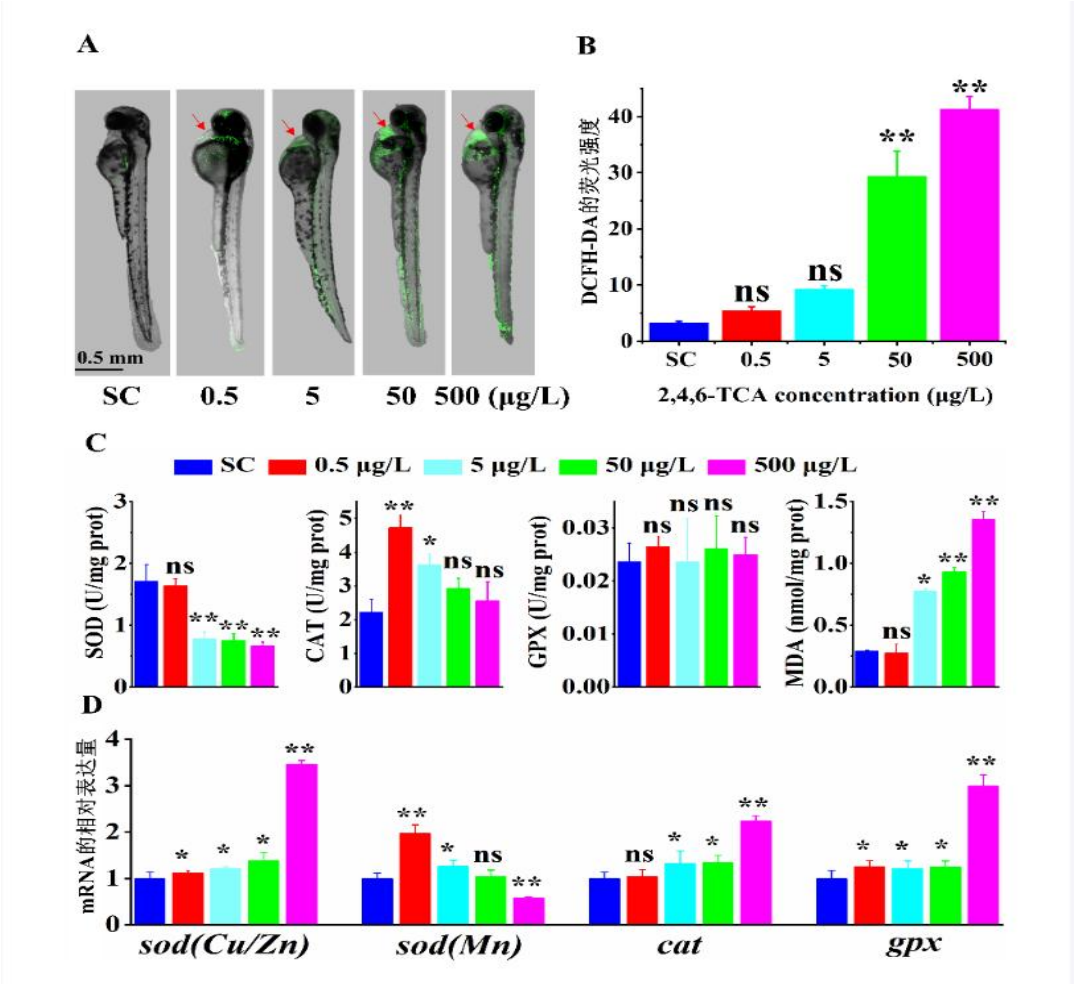
Team Research: Microplastic Exposure and Zebrafish Neurotoxicity | Experimental Design, Environmental Health

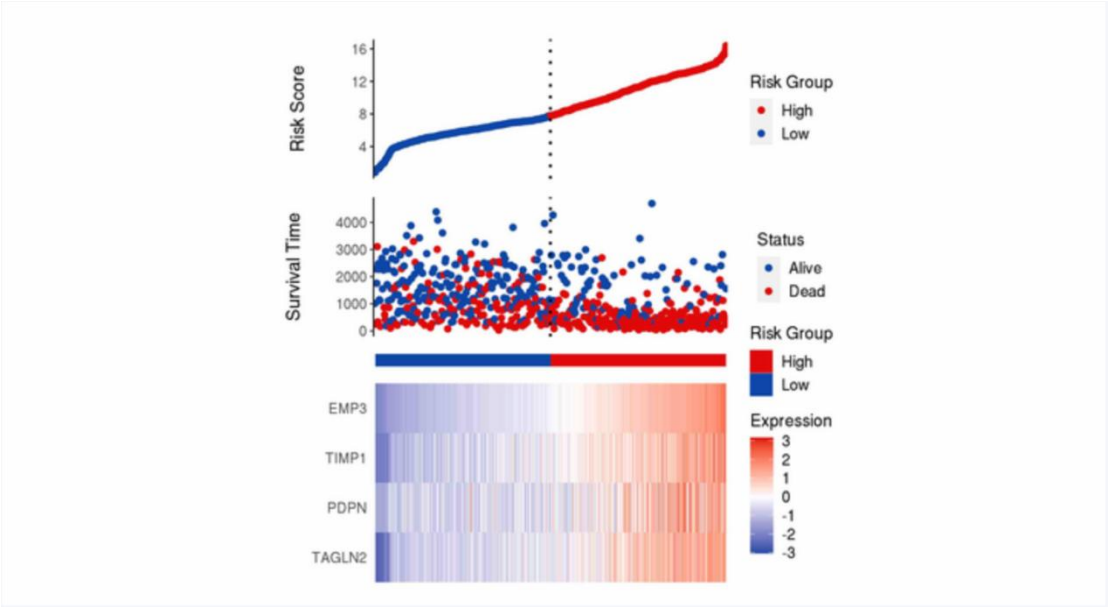
Motivated by the neurotoxic risks of microplastics across aquatic life stages, I designed a comparative toxicology study. Existing work usually examines only one developmental phase; this project compares adult zebrafish with 6-hour-post-fertilization embryos to identify stage-specific sensitivity and early markers of neural damage. I set up staged exposure groups and tracked survival rate, behavioural performance, histopathological sections, and oxidative-stress endpoints. HE and TUNEL staining were applied to detect neuronal apoptosis. Preliminary results show higher mortality among juvenile fish, with HE and TUNEL staining indicating increased neuronal apoptosis after exposure. These findings provide preliminary laboratory evidence for microplastic neurotoxicity; transcriptomic and metabolomic validation is planned to consolidate conclusions. I also presented the public-health implications at an on-campus session.




```
# pseudocode for experimental result summaryimport pandas as pd
result_df = pd.read_csv["zebrafish_toxicity_raw.csv"]
survival_stats=result_df.groupby[["life_stage","treatment"]]["survival"].agg[["mean","std
"]]
print(survival_stats]
```

Team Research: Hypertension, Remnant Cholesterol, and Arterial Stiffness | Retrospective Clinical Epidemiology

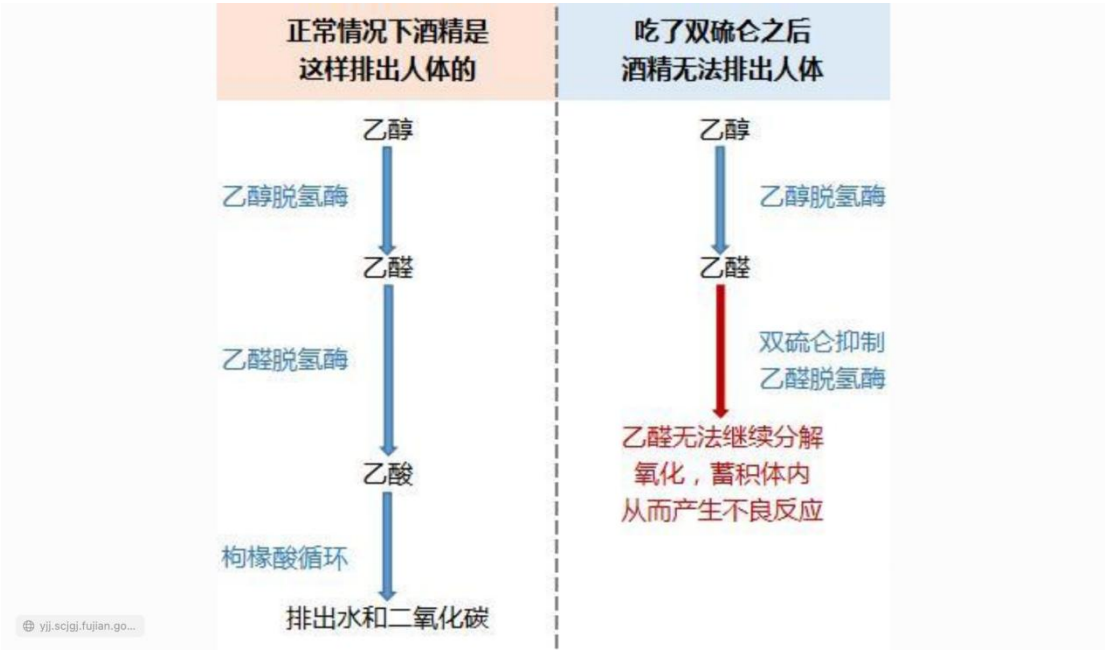
Patients often carry multiple metabolic risks simultaneously, yet simple addition of individual effects may misestimate vascular hazard. Using 4,573 adults aged 45+, this retrospective study quantifies the independent and joint effects of hypertension and elevated remnant cholesterol on arterial stiffness, and estimates the risk attributable to their biological interaction. I used brachial-ankle pulse-wave velocity, blood pressure, and lipid panel data as inputs. Multivariable logistic regression with additive interaction models adjusted for confounders and separated individual from synergistic risk. Results show that co-exposed participants have significantly higher arterial-stiffness risk ($P < 0.001$), with approximately 5%-15% of joint exposure risk attributable to interaction. The output supports clinical risk stratification and community cardiovascular prevention by reminding clinicians not to evaluate risk factors in isolation.




```
# sample analysis snippet
import statsmodels.api as sm
model = sm.Logit(y, X_confounders + X_hypertension + X_remnant + X_interaction)
result = model.fit()
print(result.summary())
```

Independent Research: Safe Medication Use and Campus Health Communication | Pharmacology, Science Outreach

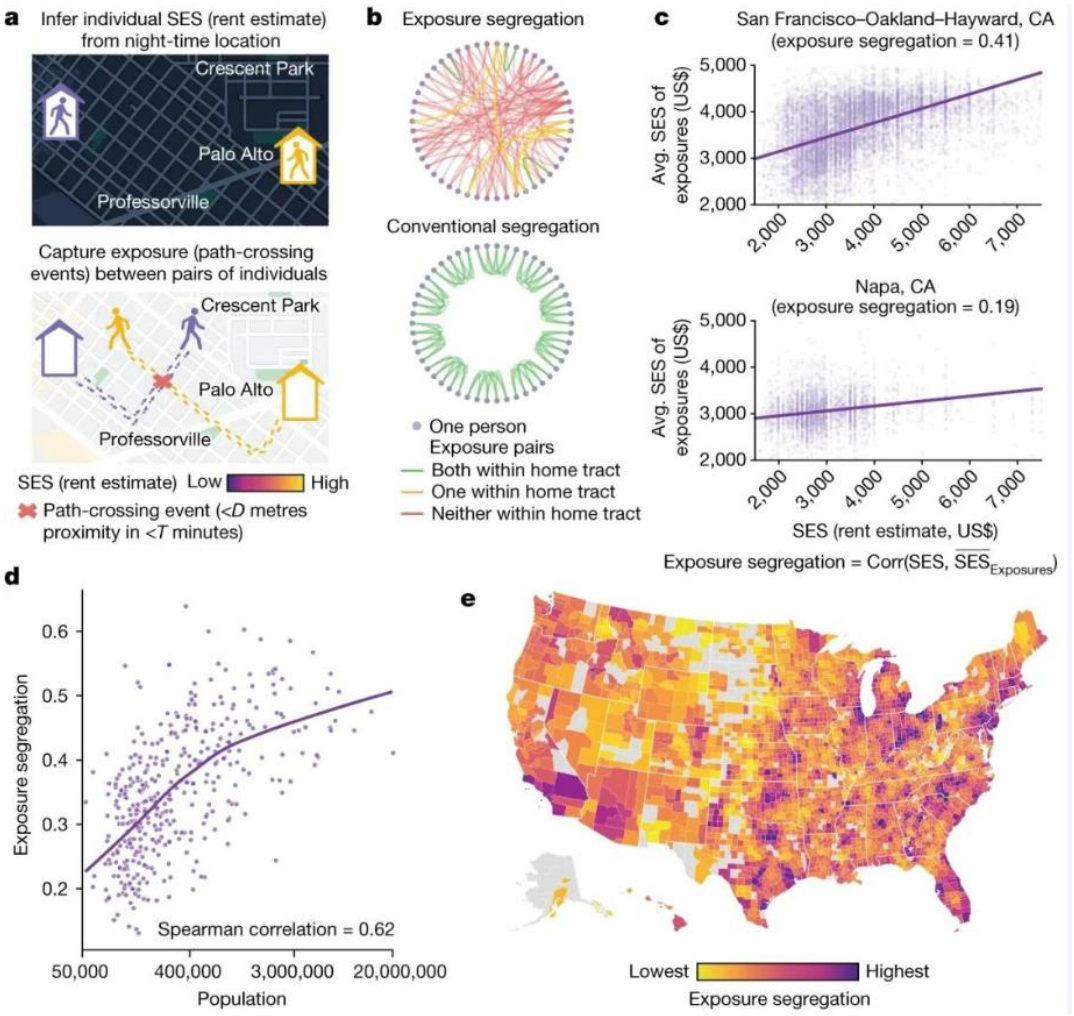
Certain cephalosporin antibiotics trigger dangerous disulfiram-like reactions when combined with alcohol, yet most students lack systematic awareness. This independent project reviews the mechanism, symptoms, and safe-use guidance of this drug-alcohol interaction, and proposes practical improvements to campus health education. I reviewed pharmacological literature and wrote an English technical report covering biochemical mechanisms, warning symptoms, and safe-medication advice. I then interviewed teaching staff to assess the current state of campus health education. I produced readable English-language science materials and presentation slides for student audiences, and collected teacher feedback to suggest concrete optimisations for campus health-promotion work. The project demonstrates how biomedical knowledge can be translated into actionable safety guidance for ordinary young people.




```
# pseudocode for feedback statistics
feedback_df = pd.read_excel("campus_health_feedback.xlsx")
suggestion_summary = feedback_df["suggestions"].value_counts()
print(suggestion_summary)
```

Independent Research: Mapping Chronic Disease Burden and Neighborhood Support in Baltimore | Public-Health Spatial Data Science

This data-driven public-health study analyzes disparities of chronic disease burden and neighborhood support constraints across 197 Baltimore census tracts. Integrating CDC PLACES health estimates, social vulnerability indicators and ACS household-income data, I constructed normalized composite scores for chronic disease burden and community support constraints, applied Moran's I for spatial autocorrelation analysis and k-means clustering to identify distinct neighborhood profiles. Using spatial-aware cross-validation, I trained ridge regression and random-forest models to predict tract-level diabetes prevalence, and performed sensitivity tests to verify the stability of high-need tract screening results. This workflow reveals geographically clustered high-risk communities and provides evidence for targeted public-health intervention.




```
# Compute percentile-rank composite burden score
def percentile_score(df, cols):
    ranks = df[cols].rank(method="average")
    u = (ranks - 1) / (len(df)-1)
    return u.mean(axis=1).to_numpy()
```